

Thermally-Weighted Interband JDOS at the X and L Critical Points of Gold

Reduction of the 3D Emission Integral to One Dimension

Following Guerrisi, Rosei & Winsemius, *Phys. Rev. B* **12**, 557 (1975)

1 Starting Point

The direct interband emission rate at photon energy $\hbar\omega$ is given by Fermi's golden rule,

$$\Gamma_e^{\text{cv}}(\hbar\omega) \propto \int_{\text{BZ}} d^3k f(\mathcal{E}_c(\mathbf{k})) [1 - f(\mathcal{E}_v(\mathbf{k}))] \delta(\mathcal{E}_c(\mathbf{k}) - \mathcal{E}_v(\mathbf{k}) - \hbar\omega), \quad (1)$$

where $f(\mathcal{E}) = [1 + e^{\mathcal{E}/k_B T}]^{-1}$ is the Fermi-Dirac distribution with energies measured from $E_F = 0$. We assume direct transitions ($\mathbf{k}_1 \approx \mathbf{k}_2$) so both bands are evaluated at the same $\mathbf{k}$.

1.1 Band Dispersions

Near each critical point the bands are expanded to second order with cylindrical symmetry ($k_\perp$ transverse to the symmetry axis, $k_\parallel$ along it):

$$\boxed{\mathcal{E}_v(\mathbf{k}) = -\mathcal{E}_{0v} - A_v k_\perp^2 - B_v k_\parallel^2, \quad \mathcal{E}_c(\mathbf{k}) = \mathcal{E}_{0c} + A_c k_\perp^2 - B_c k_\parallel^2,} \quad (2)$$

with all coefficients $A_c, B_c, A_v, B_v > 0$. The conduction band is a *saddle point* at both X and L: it curves upward along $k_\perp$ and downward along $k_\parallel$.

Table 1: Band parameters at the two critical points. Curvature coefficients $A = \hbar^2/(2m_\perp^*)$, $B = \hbar^2/(2m_\parallel^*)$ with $C \equiv \hbar^2/(2m_e) = 3.81 \text{ eV} \cdot \text{\AA}^2$.

Point	N	$\mathcal{E}_{0c}$	$\mathcal{E}_g$	Topology	Axes
X (Δ)	6	0 (ref.)	$\sim 1.94 \text{ eV}$	Saddle (M_1)	$k_\perp$: BZ face; $k_\parallel$: along Δ
L (Λ)	8	$\mathcal{E}_{0c}^L \neq 0$	$\sim 2.45 \text{ eV}$	Saddle (M_1)	$k_\perp$: hex face; $k_\parallel$: along Λ

The derivation below is **carried out once in general form**. All expressions apply identically to both critical points—only the numerical values of $(A_c, B_c, A_v, B_v, \mathcal{E}_{0c}, \mathcal{E}_g, N)$ differ.

2 Change of Variables I: Cylindrical Symmetry

Exploiting the azimuthal symmetry around the symmetry axis,

$$d^3k = dk_x dk_y dk_z \longrightarrow k_\perp dk_\perp dk_\parallel d\phi \longrightarrow 2\pi k_\perp dk_\perp dk_\parallel. \quad (3)$$

3 Change of Variables II: Linearization of Momenta

Define $u \equiv k_\perp^2 \geq 0$ and $v \equiv k_\parallel^2 \geq 0$:

$$du = 2k_\perp dk_\perp, \quad dv = 2|k_\parallel| dk_\parallel \Rightarrow dk_\parallel = \frac{dv}{2\sqrt{v}}. \quad (4)$$

The factor of 2 from the $\pm k_{\parallel}$ symmetry cancels the $1/2$, giving

$$2\pi k_{\perp} dk_{\perp} dk_{\parallel} \longrightarrow \pi \frac{du dv}{\sqrt{v}}, \quad u \geq 0, \quad v \geq 0. \quad (5)$$

In these variables the dispersions are *linear*:

$$\mathcal{E}_c(u, v) = \mathcal{E}_{0c} + A_c u - B_c v, \quad \mathcal{E}_v(u, v) = -\mathcal{E}_{0v} - A_v u - B_v v. \quad (6)$$

4 Change of Variables III: Energy Space

We introduce two physically motivated variables:

$$\mathcal{E} \equiv \mathcal{E}_c(u, v), \quad \Delta \equiv \mathcal{E}_c(u, v) - \mathcal{E}_v(u, v), \quad (7)$$

so that

$$\mathcal{E} - \mathcal{E}_{0c} = A_c u - B_c v, \quad (8)$$

$$\Delta - \mathcal{E}_g = \bar{A} u + \bar{B} v, \quad (9)$$

where

$$\boxed{\mathcal{E}_g \equiv \mathcal{E}_{0c} + \mathcal{E}_{0v}, \quad \bar{A} \equiv A_c + A_v, \quad \bar{B} \equiv B_v - B_c.} \quad (10)$$

4.1 Matrix Form and Jacobian

Equations (8)–(9) in matrix form:

$$\begin{pmatrix} \mathcal{E} - \mathcal{E}_{0c} \\ \Delta - \mathcal{E}_g \end{pmatrix} = \underbrace{\begin{pmatrix} A_c & -B_c \\ \bar{A} & \bar{B} \end{pmatrix}}_M \begin{pmatrix} u \\ v \end{pmatrix}. \quad (11)$$

The Jacobian determinant is

$$D \equiv \det M = A_c \bar{B} + B_c \bar{A} = A_c(B_v - B_c) + B_c(A_c + A_v) = \boxed{A_c B_v + A_v B_c > 0}. \quad (12)$$

Since all curvature coefficients are positive, $D > 0$ always. No sign ambiguity arises.

4.2 Inverse Transformation

$$\begin{pmatrix} u \\ v \end{pmatrix} = \frac{1}{D} \begin{pmatrix} \bar{B} & B_c \\ -\bar{A} & A_c \end{pmatrix} \begin{pmatrix} \mathcal{E} - \mathcal{E}_{0c} \\ \Delta - \mathcal{E}_g \end{pmatrix}, \quad (13)$$

giving explicitly

$$u(\mathcal{E}, \Delta) = \frac{1}{D} \left[\bar{B} (\mathcal{E} - \mathcal{E}_{0c}) + B_c (\Delta - \mathcal{E}_g) \right], \quad (14)$$

$$v(\mathcal{E}, \Delta) = \frac{1}{D} \left[-\bar{A} (\mathcal{E} - \mathcal{E}_{0c}) + A_c (\Delta - \mathcal{E}_g) \right]. \quad (15)$$

5 Integration Limits

The physical constraints $u \geq 0$ and $v \geq 0$ with $D > 0$ yield:

From $v \geq 0$ (eq. 15):

$$\bar{A}(\mathcal{E} - \mathcal{E}_{0c}) \leq A_c(\Delta - \mathcal{E}_g) \implies \boxed{\mathcal{E} \leq \mathcal{E}_{0c} + \frac{A_c}{\bar{A}}(\Delta - \mathcal{E}_g) \equiv \mathcal{E}_{\max}(\Delta)}. \quad (16)$$

From $u \geq 0$ (eq. 14):

$$\bar{B}(\mathcal{E} - \mathcal{E}_{0c}) \geq -B_c(\Delta - \mathcal{E}_g) \implies \boxed{\mathcal{E} \geq \mathcal{E}_{0c} - \frac{B_c}{\bar{B}}(\Delta - \mathcal{E}_g) \equiv \mathcal{E}_{\min}(\Delta)}. \quad (17)$$

Physical interpretation: $\mathcal{E}_{\max}$ corresponds to $v = 0$ ($k_{\parallel} = 0$, the transverse circle); $\mathcal{E}_{\min}$ corresponds to $u = 0$ ($k_{\perp} = 0$, the symmetry axis). The integration domain $\mathcal{E}_{\min} \leq \mathcal{E} \leq \mathcal{E}_{\max}$ is non-empty for $\Delta > \mathcal{E}_g$, i.e., for photon energies above the gap threshold.

6 Transformed Measure

From eq. (15), v can be expressed in terms of $\mathcal{E}_{\max}$:

$$v(\mathcal{E}, \Delta) = \frac{\bar{A}}{D}(\mathcal{E}_{\max}(\Delta) - \mathcal{E}), \quad (18)$$

which is manifestly non-negative for $\mathcal{E} \leq \mathcal{E}_{\max}$. Substituting into the measure (5) and including the $1/D$ Jacobian factor:

$$\pi \frac{du dv}{\sqrt{v}} \longrightarrow \frac{\pi}{\sqrt{\bar{A}D}} \frac{d\Delta d\mathcal{E}}{\sqrt{\mathcal{E}_{\max}(\Delta) - \mathcal{E}}}. \quad (19)$$

The integrable singularity $(\mathcal{E}_{\max} - \mathcal{E})^{-1/2}$ occurs at the **upper limit**, where $v = 0$ (the transverse circle in k -space).

7 Resolving the Delta Function

Substituting eq. (19) into eq. (1) and using $\mathcal{E}_v = \mathcal{E} - \Delta$:

$$\Gamma_e^{\text{cv}}(\hbar\omega) \propto \frac{\pi}{\sqrt{\bar{A}D}} \iint f(\mathcal{E}) [1 - f(\mathcal{E} - \Delta)] \delta(\Delta - \hbar\omega) \frac{d\Delta d\mathcal{E}}{\sqrt{\mathcal{E}_{\max}(\Delta) - \mathcal{E}}}. \quad (20)$$

The δ -function sets $\Delta = \hbar\omega$ and eliminates one integration, yielding the central result:

$$\boxed{\Gamma_e^{\text{cv}}(\hbar\omega) = \frac{\pi^2 |\mu|^2}{\hbar \sqrt{\bar{A}D}} \int_{\mathcal{E}_{\min}(\hbar\omega)}^{\mathcal{E}_{\max}(\hbar\omega)} \frac{f(\mathcal{E}) [1 - f(\mathcal{E} - \hbar\omega)]}{\sqrt{\mathcal{E}_{\max}(\hbar\omega) - \mathcal{E}}} d\mathcal{E}} \quad (21)$$

with

$$\mathcal{E}_{\max}(\hbar\omega) = \mathcal{E}_{0c} + \frac{A_c}{\bar{A}}(\hbar\omega - \mathcal{E}_g), \quad \mathcal{E}_{\min}(\hbar\omega) = \mathcal{E}_{0c} - \frac{B_c}{\bar{B}}(\hbar\omega - \mathcal{E}_g). \quad (22)$$

For absorption (ε_2), the Fermi factor becomes $f(\mathcal{E} - \hbar\omega) [1 - f(\mathcal{E})]$; at equilibrium the two are related by detailed balance $\Gamma_{\text{em}}/\Gamma_{\text{abs}} = e^{-\beta\hbar\omega}$.

8 Numerical Regularization

The singularity at $\mathcal{E} = \mathcal{E}_{\max}$ is integrable but inconvenient numerically. The substitution

$$t = \sqrt{\mathcal{E}_{\max} - \mathcal{E}}, \quad d\mathcal{E} = -2t dt, \quad \mathcal{E} = \mathcal{E}_{\max} - t^2, \quad (23)$$

transforms (21) into a smooth integral:

$$\Gamma_e^{\text{cv}}(\hbar\omega) = \frac{2\pi^2 |\mu|^2}{\hbar \sqrt{A} D} \int_0^{\sqrt{\mathcal{E}_{\max} - \mathcal{E}_{\min}}} f(\mathcal{E}_{\max} - t^2) [1 - f(\mathcal{E}_{\max} - t^2 - \hbar\omega)] dt, \quad (24)$$

which is evaluated by Simpson quadrature on a uniform t -grid.

9 Verification: Zero-Temperature JDOS

At $T = 0$ the Fermi functions become step functions, and for $\hbar\omega$ just above threshold:

$$J(\hbar\omega) \propto \int_{\mathcal{E}_{\min}}^{\mathcal{E}_{\max}} \frac{d\mathcal{E}}{\sqrt{\mathcal{E}_{\max} - \mathcal{E}}} = 2\sqrt{\mathcal{E}_{\max} - \mathcal{E}_{\min}} \propto \sqrt{\hbar\omega - \mathcal{E}_g}, \quad (25)$$

recovering the M_0 -type onset ($\sqrt{\cdot}$ threshold). Note that while each individual band is a saddle (M_1), the *energy-difference* surface $\Delta(u, v) = \mathcal{E}_g + \bar{A}u + \bar{B}v$ is a minimum in (u, v) space (both coefficients positive for $B_v > B_c$), giving M_0 onset for the JDOS.

10 Application to Gold: X vs. L Parameters

Since both critical points share the same saddle topology and identical integral structure (21), the full interband ε_2 model is simply a weighted sum:

$$\varepsilon_2(\omega) = \frac{S}{\omega^2} \left[|P_X/P_L|^2 N_X J_X(\omega, T) + N_L J_L(\omega, T) \right] + \frac{A_D}{\omega^3}, \quad (26)$$

where $J_{X,L}$ are evaluated using (24) with the respective parameters below.

The **only** differences between the X and L contributions are:

1. Different effective masses ($m_{c\perp}^*$, $m_{c\parallel}^*$, $m_{v\perp}^*$, $m_{v\parallel}^*$).
2. Different gap energies $\mathcal{E}_g$.
3. $\mathcal{E}_{0c}^L \neq 0$ (whereas $\mathcal{E}_{0c}^X \equiv 0$ by convention).
4. Different BZ multiplicities ($N_X = 6$, $N_L = 8$).

Summary of Transformations

$$\underbrace{d^3k}_{\text{3D BZ}} \xrightarrow{\text{cyl.}} \underbrace{2\pi k_{\perp} dk_{\perp} dk_{\parallel}}_{\text{2D}} \xrightarrow{u,v} \underbrace{\pi \frac{du dv}{\sqrt{v}}}_{\text{linear}} \xrightarrow{\mathcal{E}, \Delta} \underbrace{\frac{\pi}{\sqrt{AD}} \frac{d\Delta d\mathcal{E}}{\sqrt{\mathcal{E}_{\max} - \mathcal{E}}}}_{\text{energy space}} \xrightarrow{\delta} \underbrace{1\text{D}}_{(21)}$$

References: M. Guerrisi, R. Rosei, and P. Winsemius, *Phys. Rev. B* **12**, 557 (1975). N. E. Christensen and B. O. Seraphin, *Phys. Rev. B* **4**, 3321 (1971).

Table 2: Derived quantities for the two critical points. All curvature coefficients $A_\alpha = C/m_{\alpha\perp}^*$, $B_\alpha = C/m_{\alpha\parallel}^*$.

Quantity	X point	L point
Multiplicity N	6	8
$\mathcal{E}_{0c}$	0 (reference)	fitted ($\neq 0$)
$\mathcal{E}_g$	~ 1.94 eV	~ 2.45 eV
$\bar{A} = A_c + A_v$	$A_c^X + A_v^X$	$A_c^L + A_v^L$
$\bar{B} = B_v - B_c$	$B_v^X - B_c^X$	$B_v^L - B_c^L$
$D = A_c B_v + A_v B_c$	$A_c^X B_v^X + A_v^X B_c^X$	$A_c^L B_v^L + A_v^L B_c^L$
Singularity	$(\mathcal{E}_{\max} - \mathcal{E})^{-1/2}$	$(\mathcal{E}_{\max} - \mathcal{E})^{-1/2}$
$\mathcal{E}_{\max}(\hbar\omega)$	$\mathcal{E}_{0c} + \frac{A_c}{A_c + A_v}(\hbar\omega - \mathcal{E}_g)$	
$\mathcal{E}_{\min}(\hbar\omega)$	$\mathcal{E}_{0c} - \frac{B_c}{B_v - B_c}(\hbar\omega - \mathcal{E}_g)$	